

Program: Diploma in Food Processing Technology	
Course Code: 4428	Course Title: Food Chemistry I Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3(L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To determine various chemical constituents and compounds in the sample
- Demonstrate various chemical methods to evaluate chemical properties of food samples

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about food components		Fundamentals of food processing technology	II

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain various chemical methods for the detection and estimation of chemical constituents in food	8	Applying
CO2	Analyze different chemical compounds in food sample	11	Applying
CO3	Demonstrate various chemical methods to find the chemical properties of foods	15	Applying
CO4	Design, carry out, record and analyze the results of chemical experiments precisely	11	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3	3		
CO2				3	3		
CO3				3	3		
CO4				3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

CourseOutline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain various chemical methods for the detection and estimation of chemical constituents in food		
M1.01	Qualitative tests for proteins	2	Understanding
M1.02	Quantitative determination of protein	3	Applying
M1.03	Qualitative tests for carbohydrates	3	Applying
CO2	Analyze different chemical compounds in food sample		
M2.01	Detection / Estimation of some additives in foods	3	Applying
M2.02	Detection/Estimation of adulterants in some foods	3	Applying
M2.03	Determination of free fatty acid content in fats and oils	3	Applying
	Lab Test –I	2	
CO3	Demonstrate various chemical methods to find the chemical properties of foods		
M3.01	Study of some functional properties of protein	6	Applying
M3.02	Detection and estimation of oxidative rancidity in fats/oils	3	Applying

M3.03	Determination of rate/ extent of hydrolysis of sucrose/starch	3	Applying
M3.04	Study of Gelling properties of starch	3	Understanding
CO4	Design, carry out, record and analyze the results of chemical experiments precisely		
M4.01	Preparation and measurement of pH of standard buffers	3	Applying
M4.02	Determination of gluten content in wheat flour	3	Applying
M4.03	Determination of heat stability of vitamin C	3	Applying
	Lab Test– II	2	

Text/Reference

T/R	BookTitle/Author
R1	Morris B. Jacobs , The chemical analysis of foods and food products, III Edition, CBS Publishers and distributors New Delhi.
R2	ISI hand book of food analysis
R3	Ranganna S , Hand book of analysis and quality control for fruit and vegetable products, II Ed., Tata McGraw Hill Publishing Co. New Delhi.
R4	Official Method of analysis of AOAC
R5	Rui M. S. Cruz, Igor Khmelinskii, Margarida Vieira, Methods in Food Analysis 1st Edition, 2014 CRC Publishers

Online Resources

Sl.No	Website Link
1.	https://www.mpbio.com/us/oxidative-rancidity-mechanism
2.	https://www.betterhealth.vic.gov.au/health/conditionsandtreatments/food-additives
3.	https://www.vedantu.com/chemistry/buffer-solutions